

Hazem Masarani

Software Engineer — Backend, Distributed Systems & ML Infrastructure

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SUMMARY

Software engineer with experience building Python automation, data-intensive backend systems, and multi-GPU ML infrastructure. M.S. Data Science candidate with a 4.0 GPA and co-author of published research on distributed LLM inference. **EXPERIENCE**

PACELab, Stony Brook University

Graduate Researcher

Aug 2025–Present

Stony Brook, NY

- Develop tensor-parallel inference for Mamba-style state-space models across multi-GPU systems.
- Co-authored a communication-efficient design using state caching, locality-preserving partitioning, and quantized AllReduce.
- Analyze throughput, time-to-first-token, communication overhead, and scaling behavior across deployment configurations.

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Senior Data Automation & Network Data Engineer

Jan 2024–Sep 2024

Egypt (Hybrid)

- Led development of Python pipelines and backend automation processing large, multi-source telecom network datasets.
- Built live operational tooling integrating alarms, topology, rollout, availability, and maintenance signals.
- Deployed an OpenStreetMap tile server with Docker to support real-time geospatial network visualization.
- Mentored new engineers in Python, SQL, algorithms, and data-analysis fundamentals.

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Junior Data Automation & Network Data Engineer

Jul 2023–Jan 2024

Egypt (Hybrid)

- Contributed to the eSW platform, integrating multiple data sources for reliable analysis and reporting.
- Developed an internal chatbot that automated data access and routine analysis workflows.

Al-Madinah School

Computer Science Teacher

Sep 2024–Aug 2025

Queens, NY

- Taught computing across varied skill levels, translating technical concepts into clear, practical instruction.

SELECTED WORK

Scaling State-Space Models on Multiple GPUs with Tensor Parallelism

Co-author, arXiv:2602.21144 (2026). Communication-efficient multi-GPU inference for selective SSMs.

Multi-user Cloud IDE — Built an online IDE with concurrent users and secure server-side code execution; graduation project awarded A+.

EDUCATION

Stony Brook University

M.S. Data Science, GPA: 4.0/4.0

Expected Dec 2026

Stony Brook, NY

Alexandria University

B.Sc. Computer & Communication Engineering, Honors, GPA: 3.47/4.0

2017–2022

Alexandria, Egypt

TECHNICAL SKILLS

Languages: Python, Java, C/C++, JavaScript, SQL, R **Systems:** Distributed systems, parallel computing, Linux, Docker, Git, APIs

Data/ML: PyTorch, TensorFlow, Pandas, NumPy, Scikit-learn **Databases:** PostgreSQL, MySQL, SQL Server, MongoDB